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Do World of Warcraft (MMORPG) players experience less loneliness and social anxiety in online world (virtual environment) than in real world (offline)?

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ABSTRACT

The number of online game players continues to grow, as well as the number of hours spent in online world. World of Warcraft (WoW) is one of the most popular Massively Multiplayer Online Role Playing Games (MMORPG). It is set in an elaborate fictional world, in which players achieve goals through interaction and mutual communication. The present study verified differences in experiencing loneliness (UCLA Loneliness Scale) and social anxiety (Social Phobia Inventory) in WoW players in online and real world (offline) since these two worlds and the functioning of the players in them differ. In the sample consisting of 161 players, it was found that players experience a significantly lower degree of loneliness and social anxiety in online than in real world. The lower degree of loneliness experienced was also associated with playing with friends and known people, with guild membership, as well as frequent communication with teammates through VoIP (Voice over Internet Protocol) services. The results suggest that WoW is a highly social environment that encourages cooperation, communication and friendship.
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1. Introduction

In the past ten years, MMORPGs (Massively Multiplayer Online Role Playing Games) have become a favorite way of spending leisure time for young people and adults. In 2007, MMORPGs were played by nearly 50 million people around the world (according to mogdata.voig.com). MMORPG constitute a separate genre of computer games with a fully developed fictional world with acoustic and visual detail (Griffiths, Davies & Chappell, 2007). They belong to the group of multiplayer games designed primarily for computer platforms; they are considered highly social as players are intended to perform various tasks through interaction with other players, with whom they may form new relationships and friendships as in real world. Despite the fact that in the last 10 years many games of this genre have been produced, World of Warcraft (WoW) has been the most popular title, with almost 10 million active players as of 2014 (Blizzard Entertainment, Inc. 2014). Yee, 2006a (p. 325) has aptly assessed MMORPG games as follows: "they are places where people fall in love, get married, elect governors, attend poetry readings, start a pharmaceutical business, and even commit genocide. Whatever MMORPGs are, or will become, one thing is clear. They are not just games".

In relation to computer games and the Internet, it is possible to distinguish two forms of reality or versions of the world; online world (termed also virtual environment, virtual world or virtual life); and real world (real life, offline world). Virtual world according to Gilbert (2011) represents a digital environment, that has 3D graphical interface, supports massively multi-user remote interactivity, is persistent, immersive and emphasizes user-generated activities and goals. MMORPGs are „for all intents and purposes, the latest incarnation of a virtual environment in that they share the concept of a shared virtual world, a representation of the player as an avatar", characterized with egocentric perspective, stereoscopic 3-D visualization, real-time interactivity, immersion and multisensory feedback (Stanney, Hale, & Zyda, 2015, p. 12). Except some physical limitations of the online world (such as the inability to physically manipulate objects or to satisfy basic physiological needs, etc.), online world is parallel to the real world (e.g. Whang & Chang, 2004) and people behave in online world similarly as in real world (Kozlov & Johansen, 2010; Slater et al., 2006; Yee, 2006b). In the three-dimensional online world, people can experience emotions, communicate with other people, pursue their hobbies, study, travel virtually, etc. (e.g. Partala, 2011). Greater degree of freedom, anonymity and associated differences in

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communication and realization of the activities that would hardly be feasible in real world are typical for the online world. It can be concluded, that online and real world are similar but also different in number of properties. Despite the fact about the mentioned existence of two worlds, the research of experiencing of players is concentrated only in the real world. As the possibilities and qualities of the online world differ from the real world, such as the nature of social relationships and communication, it seems that some psychological characteristics should be examined separately for each environment; not in isolation; for instance, only for the real world with findings also applicable to the online world. It is also supported by a fact, that if a player is immersed in the online world, he/she experiences his/her existence in the online world, not in the real world. Therefore, the objective of the study is to verify the differences in experiencing loneliness and social anxiety in players in real and in online world. It seems that players experience less loneliness and social anxiety in online world, which is to be related to the social nature of MMORPG games.

1.1. Social aspect of MMORPGs

The most important aspect of MMORPG games is not the playing itself, but the ability to form strong friendships, in which players are often highly emotionally invested (Cole & Griffiths, 2007; Griffiths, Davies, & Chappell, 2004). Therefore, many people use MMORPGs to meet their social needs, which they are unable to satisfy in real world (Lo, Wang, & Fang, 2005). In MMORPG, players of various ages, nationalities, gender, occupation and religious beliefs meet in order to fulfill the objectives offered by the game world (Yee, 2006a). The virtual environment allows people to express themselves in a way that would otherwise be unpleasant in real world (e.g. because of their appearance, sexuality, etc.), which makes social contact much easier and comfortable thanks to a certain degree of anonymity (Cole & Griffiths, 2007; Morahan-Martin & Schumacher, 2003). Cole and Griffiths (2007) have found that two-thirds of the players form strong friendships with other players, with the mean number of good friends made within a MMORPG of seven, and more than one third of the players had met their teammates and gaming friends before in real life. However, players are actively seeking social interaction only in the virtual environment and not in real world (Kowert & Oldmeadow, 2013). Forming friendly relationships (meaningful relationships that are supportive in nature) is an important motivator for playing MMORPG (Yee, 2006a) and for a large number of players, social interactions and friendships formed in the game environment are equal to those created in real world (Williams et al., 2006).

The formation of friendly relationships is supported by the affiliation to game teams; called guilds, where players meet with a common purpose. In general, there are three kinds of guilds: PvE (Player versus Environment), PvP (Player versus Player) and RP (Role playing). Guilds are the most important part of MMORPG social life as that is where new relationships and friendships originate, which often transcend the virtual world and become parts of everyday (real) life of many players (e.g. Trepte, Reinecke, & Juechems, 2012; Williams et al., 2006). This aspect of playing was also pointed out by Domahidi, Festl, and Quandt (2014, p. 110), who found that "online gamers with a pronounced motive of searching for social capital and teamplay had a higher probability of meeting originally online friends personally". Snodgrass, Lacy, Dengah II., and Fagan (2011) believe that in players who play with their friends from the real world, game immersion is not associated with adverse effects; but instead, it reduces stress and level of MMO problematic play. Therefore, it is more important to examine players' motivation than, for instance, the length of playing or personality traits. Frequently observed adverse effects of playing computer games are loneliness and social anxiety, in reduction of which MMORPG games are thought to play a vital role.

1.2. Loneliness in online world

Loneliness as a psychological construct represents a subjectively perceived lack of satisfying social relationships; their quantity or quality, which is accompanied by discomfort and distress (Peplau, 1988). Lonely individuals are attracted to the online environment and the social interaction in it (Leung, 2011; Morahan-Martin & Schumacher, 2003); and for these individuals, this environment is an ideal social space, in which they can satisfy their need to belong. In online world, lonely people feel more like themselves; they are more open, friendlier; and they are experiencing more entertainment and share secrets with their online friends because they understand each other better (Morahan-Martin & Schumacher, 2003). Anonymity and lack of real, face-to-face communication also reduces social anxiety, which may increase the willingness to form friendly relationships. In online world, MMORPG players can also be accepted and gain prestige due to their technical skills (Morahan-Martin & Schumacher, 2003). Despite the fact that online world is attractive for lonely people, Visser, Antheunis, and Schouten (2013) did not find direct effect of playing WoW on social competence and loneliness. Kardefelt-Winther (2014, p. 122) summarizes it as follows: "Problematic real life situations can motivate a user to go online and use certain applications to fulfil unmet needs or alleviate dysphoric moods. This can have positive and negative outcomes. Positive when the compensation is successful and makes the user feel better and negative when problematic outcomes occur; these are not mutually exclusive and may coincide. When the motivations to go online are grounded in unmet real life needs or certain psychological characteristics, the risk of negative outcomes may be higher due to the intensity of use and permanence that such compensation requires".

1.3. Social anxiety in online world

Schlenker and Leary (1982, p. 642) define social anxiety as an "anxiety resulting from the prospect or presence of personal evaluation in real or imagined social situations" (such as conversation, meeting new people or public speaking). Social anxiety is also very closely linked to loneliness (Jones, Rose, & Russell, 1990), and in the form of shyness, it is also one of the factors contributing to experiencing loneliness (Peplau, 1988). Both social anxiety and loneliness represent subjectively experienced issues of interaction with other people. Social anxiety can hinder the development of friendly and romantic relationships, achieving goals at school and work, and at worst, it can develop into a severe personality disorder (Leitenberg, 1990). Virtual environment and possibilities offered by MMORPG games can solve all the adverse effects mentioned in connection to social anxiety. Like loneliness, social anxiety too, is considered by some authors to be a significant predictor of online social interaction preferences or problematic internet use (e.g. Caplan, 2007; Lee & Stapinski, 2012; Lo et al., 2005). Highly socially anxious individuals then transfer most of their social activities, including the formation of strong friendships, into the online world, where they feel safer and more comfortable than in real world. At the same time, these individuals deem themselves more successful in computer based communication than in real, face-to-face communication (Shalom, Israeli, Markovitzky, & Lipsitz, 2015) and communicate with a higher number of people online than face-to-face (Lee & Stapinski, 2012). However, according to Lo et al. (2005) online games reduce social anxiety only temporarily as they contribute to no improvements of real world social skills.

1.4. The present study

MMORPG games form a space primarily for mutual communication, co-operation and formation of new friendships, which may extend into the real world. Mutual communication does not stay at the level of written text but is also promoted by the use of webcams and VoIP communication programs such as TeamSpeak or Ventrilo, which enable players to see and hear each other. In the virtual environment, which is often preferred to the real environment, players feel more comfortable, safer and are willing to form friendships and confine to others, therefore we assume that players experience less loneliness and social anxiety in online world than in real world. Following hypotheses are proposed with regard to differences expected in experiencing loneliness and social anxiety in online and real world:

H1. Players are expected to experience significantly less loneliness/social anxiety in online world than in real world.
H2. Guild members are expected to exhibit significantly less loneliness/social anxiety than non-members in online world only.
H3. Players who play with friends/people known from the real world are expected to exhibit significantly less loneliness/social anxiety than players who do not play with friends/known people both in online and real world.
H4. Players who have played WoW for a longer amount of time are expected to exhibit significantly less loneliness/social anxiety than players who have played for a shorter amount of time.
H5. Players who use VoIP services to communicate with other players more frequently are expected to exhibit significantly less loneliness/social anxiety than players who do not use VoIP services.

2. Method

2.1. Participants

Players found on World of Warcraft internet message boards were asked to kindly participate in research. A total of 180 players responded, out of which 19 responses were incomplete and had to be discarded. Missing values were handled with Expectation-Maximization (EM) method. The final sample consisted of 161 people, out of which 142 were male and 19 were female, and more than two thirds of the players were from the United States. The average age of the sample was 21; the minimum age was 13; and the maximum age was 50. 124 players were guild members, while 36 players were not affiliated with any guild. 112 players played with friends/people known from the real world, while 49 players played without friends/people known from the real world. The average amount of time played per week was 20.77 h, with the minimum of 3 h and maximum of 65 h per week. The average total time played was 190.56 days, with the minimum of 2 days and the maximum of 745 days. With regard to the frequency of communication programs used by players to communicate with each other, it was found that 21 players never use the communication programs; 20 players reported that they used them rarely; 45 players used them sometimes; and 74 players reported that they used them always.

2.2. Procedure

Questionnaires assessing loneliness and social anxiety were administered simultaneously two times; once for the real world and once for the online world. While completing the questionnaires, players were instructed to imagine that the questions related solely to either the real world or the online world. (Example for SPIN1: In this part, you are going to answer 17 short assertions. This entire section is focused on real world situations only. Please, answer every assertion honestly).

2.3. Measures

The questionnaire completed by the participants included sections assessing loneliness, social anxiety, demographics, game experience and game behaviors.

Loneliness was assessed using the UCLA Loneliness Scale (Russel, Peplau & Ferguson, 1978). This scale includes 20 Likert-type questions on a four-point scale, with 1 = I never feel this way and 4 = I often feel this way. Loneliness was determined as the total score of responses to the total of 20 questions. The mean score for the real world was 38.14 (SD = 14.25). The mean score for the online world was 35.00 (SD = 13.89). The questionnaire reliability expressed as the value of coefficient omega in real world was 0.95, 95% CI [0.94, 0.96]; in online world 0.95, 95% CI [0.93, 0.96].

Social anxiety was assessed using the Social Phobia Inventory SPIN (Connor, Davidson, Churchill, Sherwood, Foa & Wesler, 2000), which was designed to assess symptoms specific to social anxiety disorder. SPIN was chosen because it consists of items applicable for both the real and online world situations. The questionnaire consisted of 17 items, which measured (a) the fear in social situations (six items), (b) the avoidance of performance or social situations (seven items), and (c) the physiological discomfort in social situations (four items). Individuals were asked to respond to questions regarding how much they were bothered by particular symptoms during the past week. Each item was measured on a 5-point Likert scale, ranging from 0 (not at all) to 4 (extremely). The mean score for the real world was 34.90 (SD = 13.15). The mean score for the online world was 24.59 (SD = 9.15). The questionnaire reliability expressed as the value of coefficient omega in real world was 0.92, 95% CI [0.90, 0.94], in online world 0.91, 95% CI [0.87, 0.95].

Within the game experience and game behaviors, the following items were being assessed: affiliation to a guild, average time played per week, total time played (the total time spent playing in days), the frequency of communication with teammates through communication programs (How often do you talk through communication software? such as TeamSpeak or Ventrilo) and playing with friends/known people (Do you often play with a known associate of yours? such as friends, colleagues, classmates, neighbors, etc.).

3. Results

A paired-samples t-test was conducted to compare players' loneliness and social anxiety in online and real world conditions. Significant differences were found in the scores for loneliness in online world (M = 35.00, SD = 13.89) and real world (M = 38.13, SD = 14.25) conditions, $t(160) = -2.85$, $p = 0.005$. The effect size for this analysis ($d = 0.22$) was found to have approached the convention for a small effect (Cohen, 1977). Significant differences were found in the scores for social anxiety in online world (M = 24.59, SD = 9.15) and real world (M = 34.90, SD = 13.15) conditions, $t(160) = 10.9$, $p = 0.000$. The effect size for this analysis ($d = 0.88$) was found to have approached the convention for a large effect (Cohen, 1977). The effect size was determined using the formula t_c described by (Dunlop, Cortina, Vaslow, and Burke 1996). The results of the paired-samples t-test along with means and standard deviations for loneliness and social anxiety for both conditions are presented in Table 1.

Since social anxiety variables in online and real world were positively skewed, bootstrapping with 10 000 bootstrap samples

Table 1
Results of t-test and descriptive statistics for lonesliness and social anxiety by type of a world.

	n	M	SD	95% CI	95% Bootstrap CI	t	df	*d*
Loneliness in online world	161	35.00	13.89	[−5.29, −0.96]		-2.85^{***}	160	0.22
Loneliness in real world	161	38.13	14.25					
Social anxiety in online world	161	24.59	9.15	[−12.18, −8.44]	[-12.17, −8.51]	-10.9^{***}	160	0.88
Social anxiety in real world	161	34.90	13.15					

Note. $^{***}p < 0.01$.

was used. The result was found significant at the level of $p < 0.01$.

A mixed between-within subjects analysis of variance was conducted to compare scores for social anxiety and loneliness between 1) players who play with known people (like friends, colleagues, classmates, neighbors) and players who do not; 2) members and non-members of guilds; 3) players who communicate with other players through communication software with various frequency (never, rarely, sometimes, always); 4) players who have different total time played (2–62days, 67–148days, 150–295days, 300–745days).

3.1. Playing with known people

No significant interaction was found between the type of the world (online and real) and playing with known people for loneliness, $F(1, 159) = 1.323$, $p = 0.252$, and for social anxiety, $F(1, 159) = 0.227$, $p = 0.634$. a) The main effect of the type of the world was found significant for loneliness, $F(1, 159) = 4.746$, $p < 0.05$; $\eta^2 = 0.029$. This effect reveals that players perceive loneliness as significantly lower in online world ($M = 35.00$, $SD = 13.89$) than in real world ($M = 38.13$, $SD = 14.25$). The main effect of the type of the world was also found significant for social anxiety, $F(1, 159) = 104.087$, $p < 0.001$, $\eta^2 = 0.396$. Players perceive anxiety as significantly lower in online world ($M = 24.59$, $SD = 9.15$) than in real world ($M = 34.90$, $SD = 13.15$). b) The main effect of playing with known people is not presented as comparing average scores for loneliness and social anxiety for both worlds is not relevant.

Instead, a one-way between subjects ANOVA was conducted to compare the effect of playing with known people separately for each of the variables mentioned. A significant effect was found for loneliness in online world, $F(1, 159) = 10.708$, $p = 0.001$, $d = 0.56$ and offline world, $F(1, 159) = 3.970$, $p = 0.048$, $d = 0.34$. This effect reveals that players playing with known people perceive loneliness as significantly lower in online world ($M_1 = 32.70$, $SD_1 = 11.81$) and in real world ($M_2 = 36.66$, $SD_2 = 13.22$) than players not playing with known people ($M_1 = 40.26$, $SD_1 = 16.72$; $M_2 = 41.48$, $SD_2 = 16.00$). No significant effect was found for social anxiety in online world, $F(1, 159) = 0.987$, $p = 0.322$ and offline world, $F(1, 159) = 1.273$, $p = 0.261$.

3.2. Playing in a guild

No significant interaction was found between the type of the world (online and real) and being a member of a guild for loneliness, $F(1, 158) = 1.067$, $p = 0.303$. a) The main effect of type of the world is presented above. b) A one-way between subjects ANOVA was conducted to compare the effect of membership in a guild for loneliness. Significant effect of membership in a guild was found for loneliness in online world, $F(1, 158) = 8.511$, $p = 0.004$, $d = 0.55$. Members of a guild experienced less loneliness ($M = 33.12$, $SD = 12.69$) than players who are not members of a guild ($M = 40.47$, $SD = 15.27$). No significant effect of membership in a guild was found for loneliness in real world, $F(1, 158) = 3.067$, $p = 0.082$.

3.3. Communication with other players

Significant interaction was found between the type of the world (online and real) and communication with other players for loneliness, $F(3, 157) = 4.588$, $p = 0.004$, $\eta^2 = 0.08$. The effect size for this analysis ($f = 0.29$) was found to have approached the convention for a medium effect (Cohen, 1977). The interaction is presented in Fig. 1.

As can be seen in Fig. 1, players who communicate with other players through communication software such as TeamSpeak or Ventrilo rarely or not at all experience more loneliness than players who communicate with other players always or frequently. These differences are most noticeable in online world, where loneliness of the players who communicate with other players sometimes and always is lower than in real world. Means and standard deviations are presented in Table 2.

No significant interaction was found between the type of the world (online and real) and communication with other players for social anxiety, $F(3, 157) = 1.123$, $p = 0.342$. a) The main effect of the type of the world is presented above. b) A one-way between subjects ANOVA was conducted to compare the effect of different frequency of communication for social anxiety. Significant effect of communication with other players was found for social anxiety in online world $F(1,157) = 5.249$, $p = 0.002$, $f = 0.31$. Post hoc comparisons using the Hochberg's GT2 test indicated that players who communicate with other players always ($M = 22.55$, $SD = 6.54$) and sometimes ($M = 23.77$, $SD = 6.42$) experienced less social anxiety than players who do not communicate at all ($M = 30.09$, $SD = 16.65$). No significant effect of communication with other players was found for social anxiety in real world $F(1,157) = 2.039$, $p = 0.111$.

3.4. Different total time played

Four groups of players who had played for different total time were divided into quartiles. No significant interaction was found between the type of the world (online and real) and total time played ((2–62days, 67–148days, 150–295days, 300–745days) for social anxiety, $F(3, 112) = 2.464$, $p = 0.066$ and for loneliness, $F(3, 112) = 1.034$, $p = 0.380$. a) The main effect of the type of the world was found significant for social anxiety, $F(1, 112) = 94.88$, $p = 0.000$, $\eta^2 = 0.46$. The main effect of the type of the world was also found significant for loneliness, $F(1, 112) = 9.188$, $p < 0.001$; $\eta^2 = 0.08$. b) A one-way between subjects ANOVA was conducted to compare the effect of different total time played for social anxiety and loneliness. No significant effect was found for social anxiety in online world, $F(3, 112) = 0.644$, $p = 0.588$, for social anxiety in offline world, $F(3, 112) = 2.593$, $p = 0.056$, for loneliness in online world, $F(3, 112) = 0.903$, $p = 0.442$ and for loneliness in real world, $F(3, 112) = 2.270$, $p = 0.084$.

4. Discussion and conclusions

Every player of MMORPG games lives what it seems to be a double life; one in the real world and the other in the world of

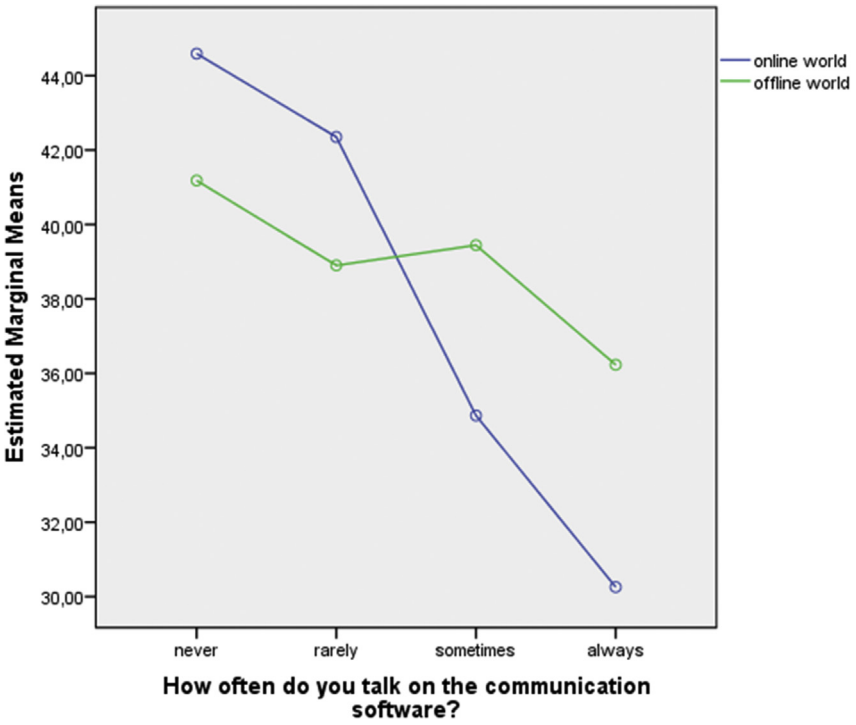


Fig. 1. Interaction between the type of the world (online and real) and communication with other players in loneliness.

Table 2
Descriptives for communication with other players.

How often do you talk on the communication software								
	Never (n = 21)		Rarely (n = 20)		sometimes (n = 45)		Always (n = 74)	
	M	SD	M	SD	M	SD	M	SD
online	45.33	15.16	42.35	14.77	34.87	12.25	30.26	12.00
real	42.09	16.38	38.90	14.81	39.44	13.89	36.22	13.61

Azaroth. The nature of the online world enables players to form friendships and communicate with others more easily. Generally, the social aspect of MMORPG games may be related to different mental experiencing in online world. This reasoning was the basis for the objective of the present study; i.e. to verify the differences in experiencing loneliness and social anxiety in players in real and online world. The results of the study support the hypothesis that World of Warcraft players experience less loneliness and social anxiety in online world. The degree of loneliness and social anxiety experienced in online world is also reduced by the factors that increase the degree of social activity such as presence of and playing with known people, playing with a team-guild (valid only for loneliness) and communication with other players. The total time played was not found significant for differences in observed variables.

Why do players experience less loneliness and social anxiety in online world than in real world? In online world, WoW players meet people who share their interests and experiences with computer games, with whom they can discuss them. Online environment provides players, who are often anxious, with a space, in which they can form and experience high-quality relationships, which satisfy their need to belong; for example, also by being members of various guilds (e.g. Cole & Griffiths, 2007; Leung, 2011). Many players use MMORPG to meet their social needs, which they are unable to satisfy in real world (Lo, Wang & Fang, 2005) as social interactions and friendships formed in game environment are equal to those in real world (Williams et al., 2006). Similarly, found that WoW players who communicate with a number of other players feel less lonely; and at the same time, this communication improves their social well-being. In contrast, players may not feel accepted in real world and in company of people who do not play or approve of computer games. According to Erath, Flanagan, and Bierman (2007), this rejection may cause social anxiety, and compounded by the lack of friends, with whom they could share their interests, also loneliness (e.g. Weiss, 1973). Players, unable to find satisfying social contacts in real world, thus turn to virtual world; and in this way, they not only experience less loneliness, but by being accepted in online world, also less social anxiety. Another explanation may be related to the perception of social anxiety as „a response to threats to social status or reputation“ (Nesse, 1998, in Crozier & Alden, 2001, p. 4). In the online environment, WoW players may perceive less threat to their social status, which can in turn be increased by their game achievements (e.g. Morahan-Martin & Schumacher, 2003). Yen, Yen, Chen, Wang, Chang, & Ko (2012) have found similar differences in social anxiety between online and real world; although not directly for online games.

Why do players playing with known people experience less loneliness both in online and real world than players not playing with known people? Why do players playing in guilds experience less loneliness in online world than players playing by themselves? Why do players who communicate with others through the communication software more often experience less loneliness and social anxiety than players not communicating as often? For beginners, the social environment of WoW may be completely new; for experienced players it may be, for example, joining a new guild. The presence of a known person or a friend is expected to help players reduce their anxiety and loneliness since they would feel more secure when not alone (cf. Snodgrass, Lacy, Dengah II., and Fagan, 2011). Similarly as to when students first come to school,

they are in a class full of strangers, so are players who are not members of guilds and play WoW by themselves; they have fewer opportunities for communication than members of guilds since the nature of playing in a guild requires constant contact and interaction (e.g. Zhong, 2011). Playing in a guild or e-Sports clan helps players satisfy their need to belong (e.g. Cole & Griffiths, 2007; Martončik, 2015), since "… playing in a guild or clan, being involved in its management and participating in offline events fosters communication with fellow players, enhances the willingness for self-disclosure and thereby increases the chances of gathering social capital" (Reer & Krämer, 2014, p. 187). Being a member of a WoW guild is typically associated with higher levels of social support (Longman, O'Connor, & Obst, 2009; Zhong, 2011). Internet chatting alone was shown by Shaw and Gant (2002) to reduce loneliness and increase perceived social support. Interaction in WoW is even more intensive and VoIP service is used for various modes of communication, not just for typing, e.g. on own chat channels, which guild members typically have. It is exactly VoIP services, which enable a higher-quality form of communication and interaction; and that supports friendship development, which can reduce perceived loneliness (e.g. van Rooij, Schoenmakers, van den Eijnden, Vermulst, & van de Mheen, 2014). WoW thus becomes an ideal online environment for forming relationships and communication with other people, which is also stressed by Williams, Ducheneaut, Xiong, Zhang, Yee, and Nickell (2006, p. 357): "for players who knew each other beforehand, WoW was an important way for them to maintain and even reinforce their relationships. For most others, it was an entrée to bridging social capital that could build up into something more over time—ranging from a few weeks to a year. For most, this was akin to a mild form of bonding found in real-world third places. Still, only a handful of players felt that these relationships mattered more than "real"-life ones".

Why do players who have played games for different total amout of time not differ in experiencing loneliness and social anxiety in online and real world? Based on the fact that the frequency of playing per week may vary throughout the weeks, the total time played, as data that is not/does not have to be estimated since it is accurately displayed by some add-ons or in-game commands (/played time), is a better indicator of the time spent by playing. At the same time, it creates a more accurate profile of players, since for instance, new players, for whom everything in the game is new and unknown may play an average of 30 ha a week; however, if they have played only for 2 weeks, their profile as MMORPG players is disputable. On the other hand, there is no doubt that players who have spent a total of 190 days in online world are MMORPG players. Why, then, is the total amount of time spent by playing not related to experiencing loneliness and social anxiety? With regard to human relationships, the online world is probably quite similar to the real world. Various groups, friendships and relationships are created in online world in a similar way as in real world. The fact that players have spent some time in a certain environment may not necesarily mean that they do not feel lonely or anxious in this environment. It is possible that during this time, players have formed several friendly relationships and joined a number of guilds. It works similarly in real world; people are members of certain groups, such as classrooms or working groups, in which they form relationships with other people; however, it often happens that people in these groups change. It is also necessary to take into account that players play games other than just WoW, and thus it is possible that they made friends, with whom they play other multiplayer games in addition to WoW. Therefore, the total time played may not correspond to the number of satisfying and accepting relationships in a sense, the more time spent in a certain online environment, the less loneliness players perceive. Sometimes it is not as important how much time players spend in a certain environment as what people surround them, what groups they are members of; basically, what kind of environment they are in. The total time played may thus not necessarily be the determining factor for experiencing loneliness. Different results were found by e.g. Lo, Wang and Fang (2005), in whose research the heaviest players of online games had the least satisfying interpersonal relationships, less than light users and non-players; and in Wei, Chen, Huang, and Bai (2012) research, they tended to have more severe social phobia symptoms, which may be related to using the average time played per week/day instead of the total time played, as mentioned above, and not specifying the type or genre of online games played.

Limitations. As interactions were not found to be significant, the main effects should be interpreted with caution; however, they are consistent with our hypothesis. The internal validity of the study could have been compromised by the fact that some aspects were not measured or observed, such as the type of the guild (e.g. PvE, PvP, RP), reasons why some players are not members of a guild or whether they were members in the past, and players' motivation for guild membership or for playing itself. It should also be noted that only one MMORPG game was examined and the data obtained through the internet message boards may not be representative of other players since players who visit these boards may represent a separate subgroups of WoW players.

The objective of the study was not to examine differences in loneliness and social anxiety between the players and non-players, but to explain players' mental experiencing in two distinct environments, in online and in real world. In our point of view, the existence of two different worlds haven't been considered in exploration of experiencing of players, yet. The results of our study refer a decrease in a social anxiety and loneliness in online world. The existing research has already provided some answers to questions related to clarifying the causes (e.g. Leung, 2011; Lo, Wang & Fang, 2005; Morahan-Martin & Schumacher, 2003; Yee, 2006a) and consequences (e.g. Lo, Wang & Fang, 2005; Kowert & Oldmeadow, 2013; Visser, Antheunis, and Schouten, 2013) of found differences between two worlds. But other questions are appeared, for example: What is the nature of changes in experiencing loneliness and social anxiety of new players? As a theme for the further research may be reflections about a lower level of loneliness in online environment, which, based on its definition, is related to the development of friendly relations in this environment, however, the question is if a creation and development of these relationships in the online environment leads to the development of social skills, which the player can later use in a creating new relationships in the real environment. Following this, we present our reflection about the search of alternatives to the MMORPGs in specific cases, when MMORPGs function as a substitution of a real world for a satisfaction of particular human needs. MMORPG games and online games in general are usually viewed in black-and-white terms; either as undesirable and having negative influence on players or as possibly carrying certain benefits. However, evaluations of MMORPG games should be presented together with other significant considerations; for example, if MMORPG games/online gaming are evaluated as negative, it is necessary to add some suggestions such as; what steps should be taken in order to motivate individuals not to become players and play for 50 h a week, or how and what alternative should be created that would be equally attractive to online game environment, which would make players feel comfortable and enable them to make friends and communicate with others easily. Especially for socially unskilled individuals (Kowert & Oldmeadow, 2013) communication with a large number of partners in the virtual environment and online gaming itself is beneficial for socialization (Kowert & Oldmeadow, 2015; Visser, Antheunis & Schouten, 2013). Many friendships

made online then expand into the real world (Domahidi, Festl, & Quandt, 2014). As noted by Kowert, Vogelgesang, Festl, and Quandt (2015, p. 56), "it can be concluded that online video games exposure and non-problematic engagement is not a risk factor for low self-esteem, low sociability, or increased loneliness for adolescent or adult players".

Following is a discussion offered by the authors of the present study. We think, that one of the main benefits of MMORPG is that it may act as a form of loneliness intervention. Luhmann, Schönbrodt, Hawkley, and Cacioppo (2014) state that loneliness intervention literature addressed four main types of interventions: (1) enhancing social skills, (2) providing social support, (3) increasing opportunities for social interaction, and (4) addressing maladaptive social cognition. MMORPG is believed to address the first three types of intervention. Speaking about the alternative to online gaming, it is necessary to understand players' motivation to play. If the online environment acts only as a substitute for the real environment, then the activities focused on social skill development and loneliness, and social anxiety treatment and prevention seem to be beneficial. It may be useful to mention a case study conducted by King, Valenca, Silva, Baczynski, Carvalho, and Nardi (2013), in which a social phobia patient used computer games and online world environment as a means of forming relationships and maintaining communication with other people; and once the patient was successfully treated with medication and CBT treatment, the time he spent in online world has decreased while his exposure to real-life situations has increased. If players are motivated to play just for fun, an alternative is not needed as computer games can be a way of spending leisure time just like making paper models or playing table tennis.

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